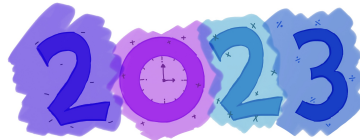


The 2023 Puzzle

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1 The 2023 Puzzle

Use all of the digits in 2023 (2,0,2,3) to create expressions that result in the numbers from 1-30.

1.1 Rules

Use arithmetic operations (addition, subtraction, multiplication, division), factorial, exponentiation, and/or concatenation (e.g., $52 = 22 + 30$, but not $(\times * 2 + 0)3 = 43$) to find as many solutions as possible. Use each digit only once.

Example: To produce 1, you can do $(2 \times 0) + (3 - 2) = 1$

1 =
2 =
3 =
4 =
5 =
6 =
7 =
8 =
9 =
10 =
11 =
12 =
13 =
14 =
15 =
16 =
17 =
18 =
19 =
20 =

21 =
22 =
23 =
24 =
25 =
26 =
27 =
28 =
29 =
30 =